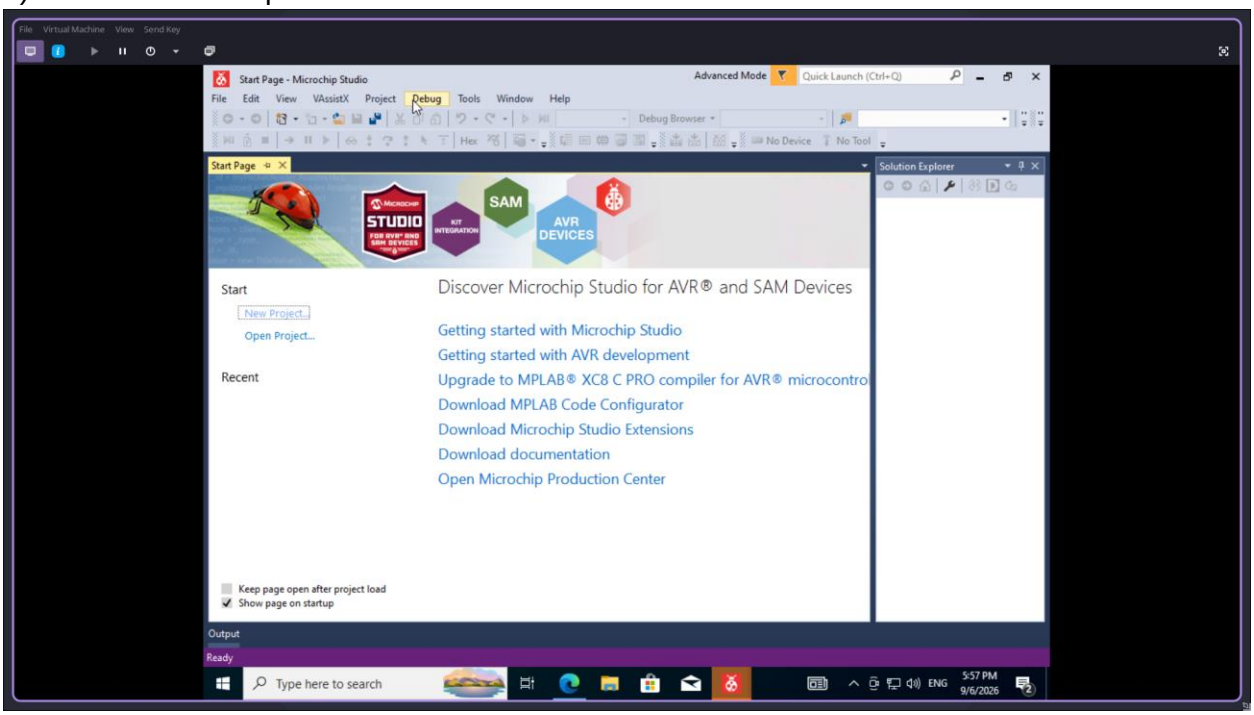
dFaculty of Computer Technology and Cybersecurity
Department of Computer Engineering
ARCHITECTURE AND ORGANIZATION OF COMPUTER SYSTEMS
LABORATORY WORK - 6

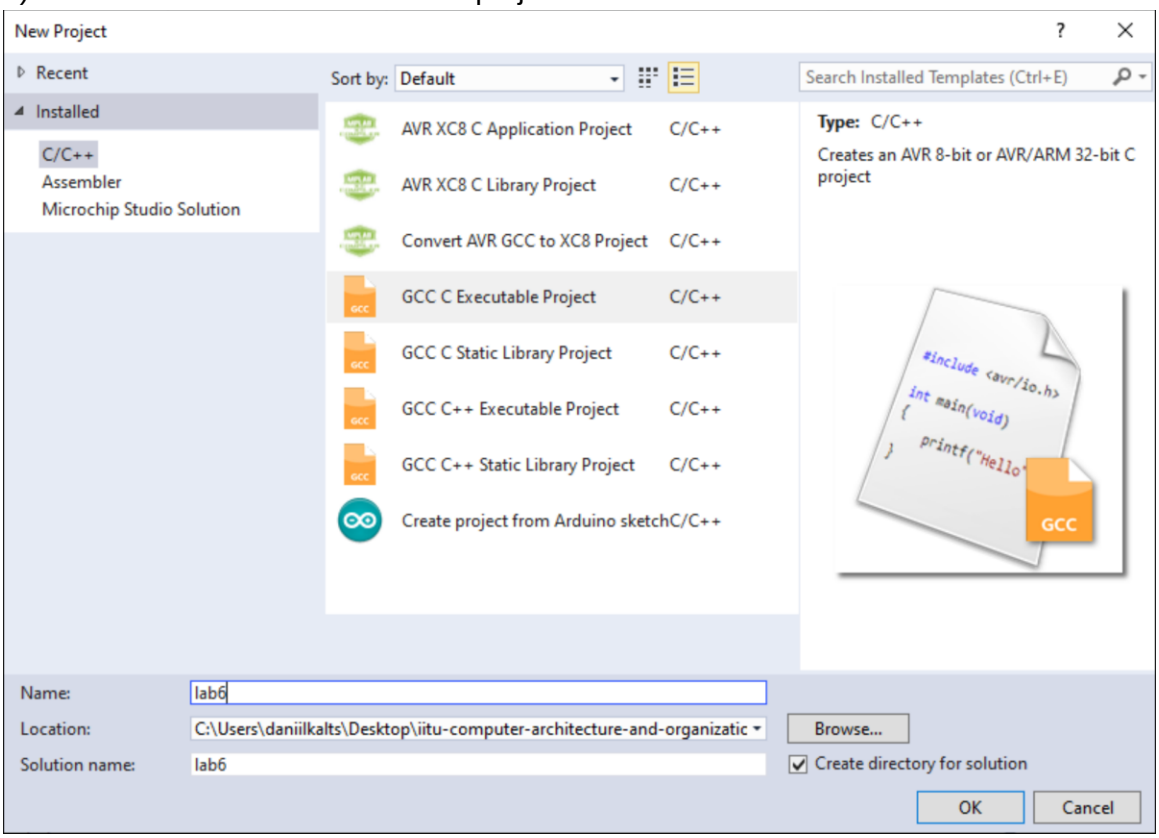
Date: 08.11.2026
Full Name: Daniil Kalts
Group: IT2-2404SE

PROGRESS OF WORK

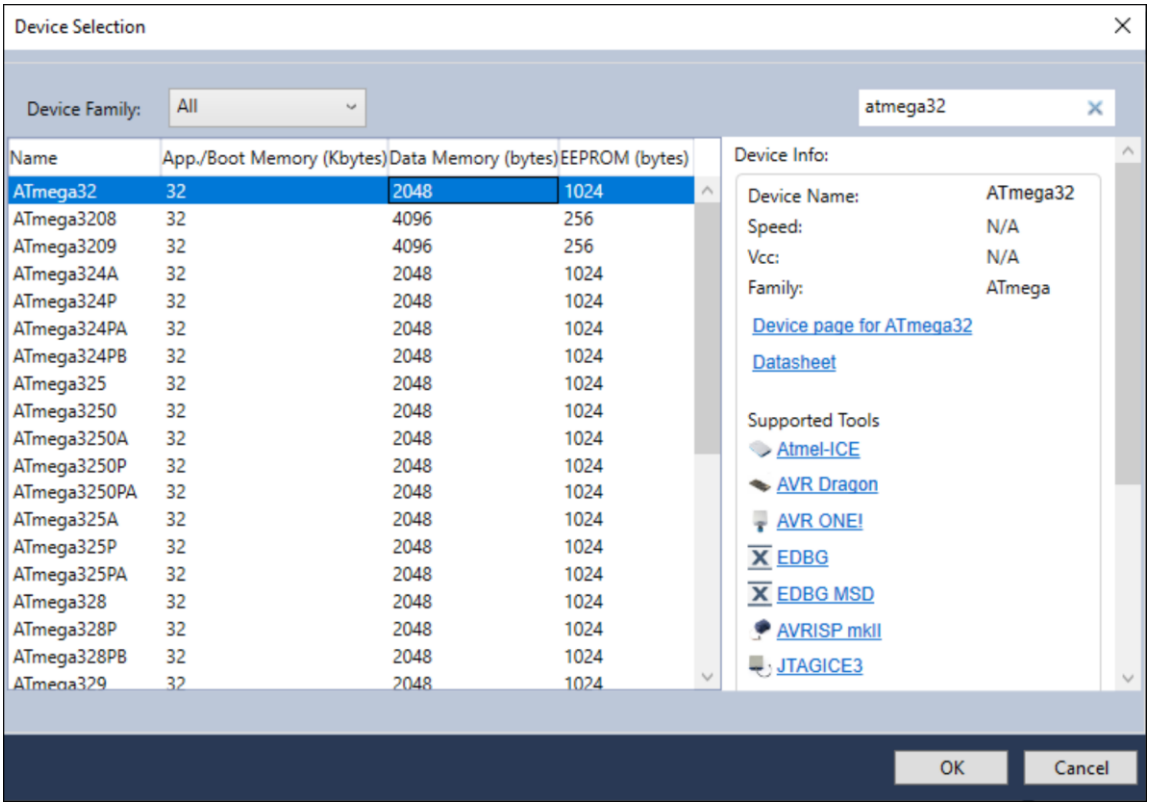
1) Start the Microchip Studio.



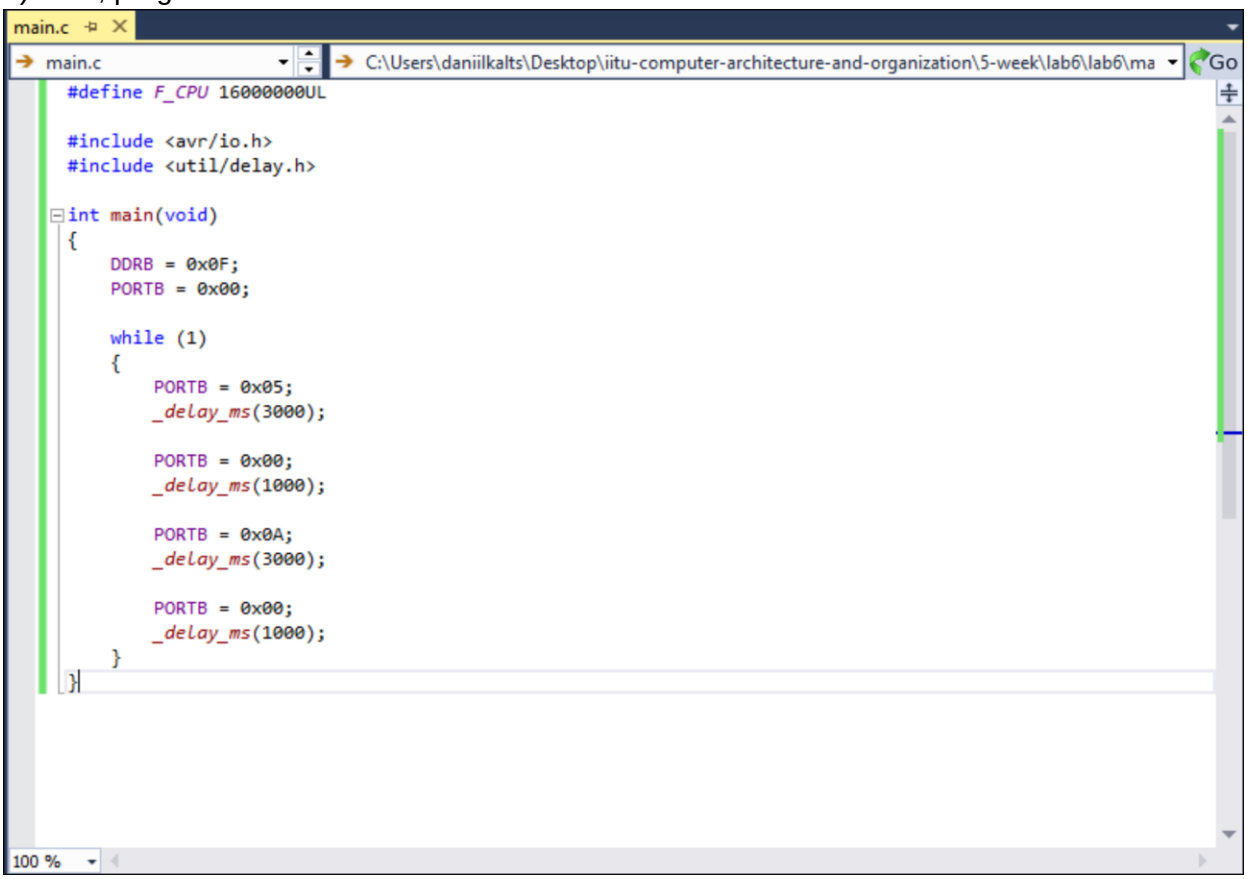
2) Create a new GCC C Executable project.



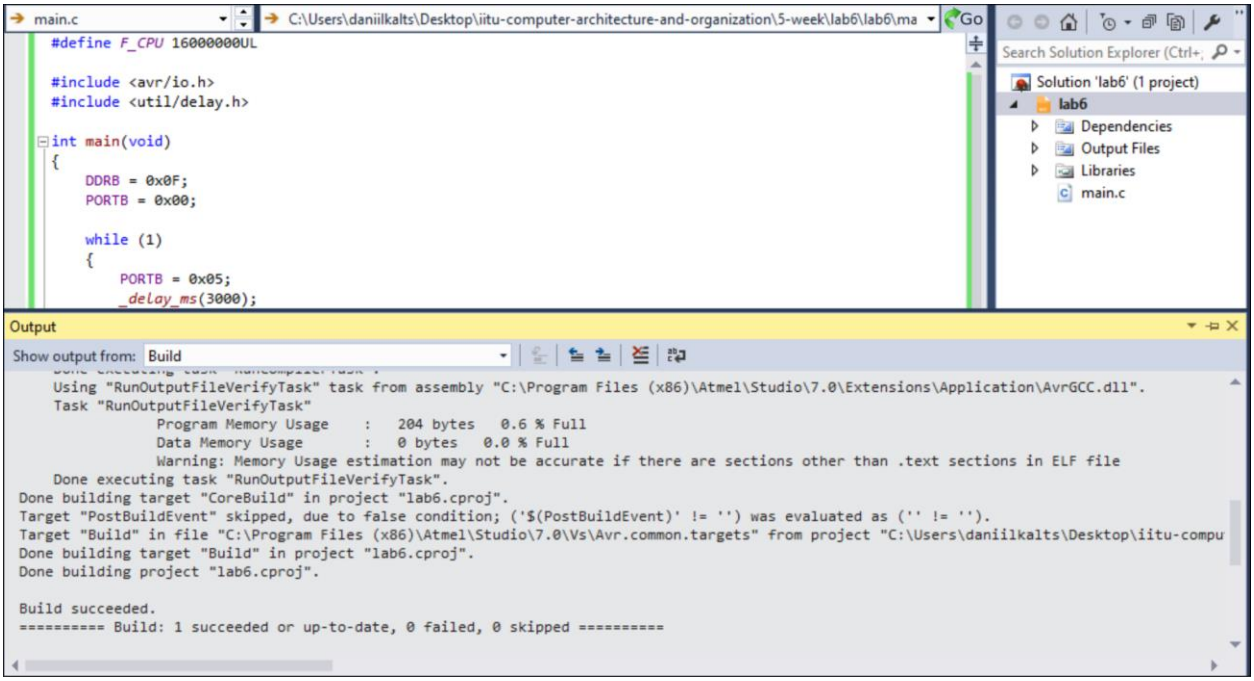
3) Select the device (in this work — ATmega32).



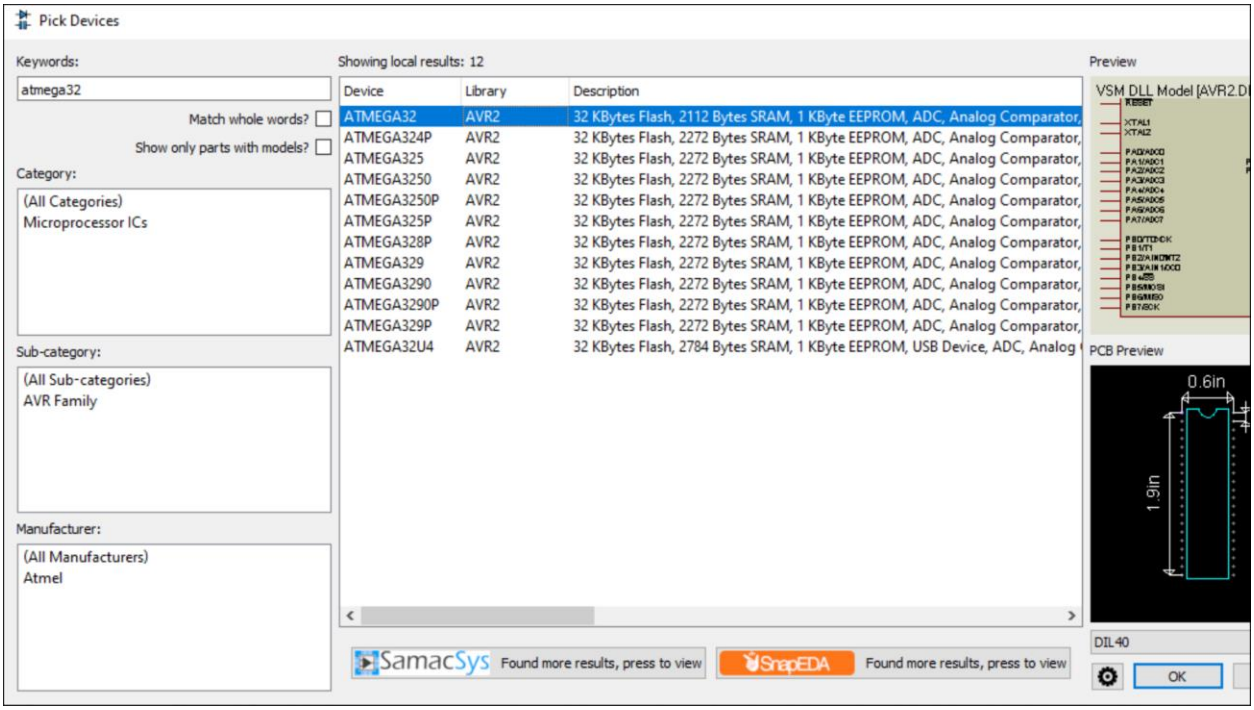
4) Next, program the microcontroller.

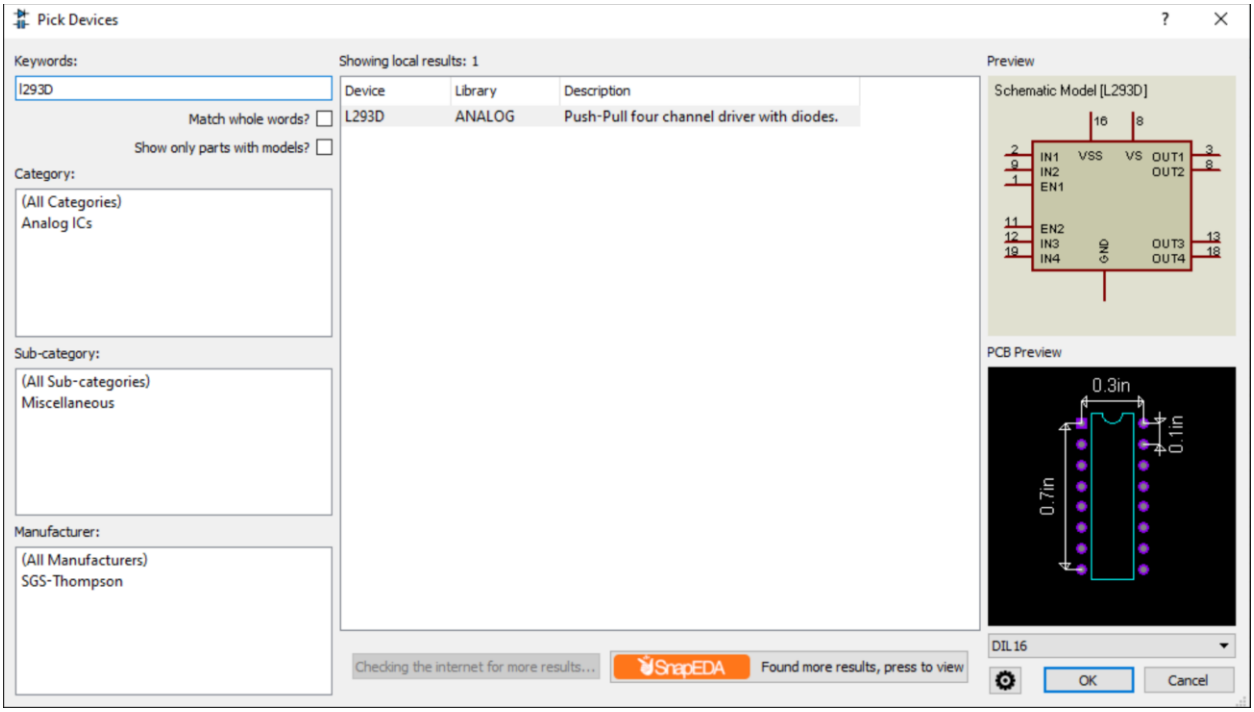


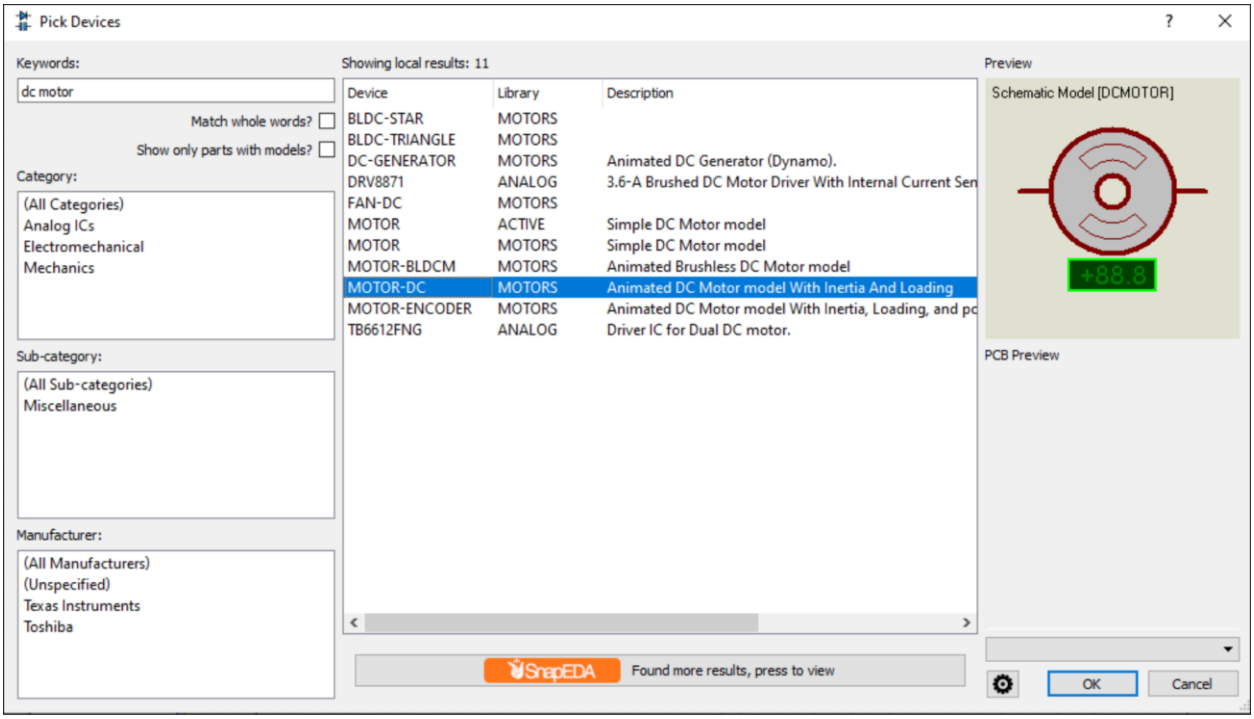
5. Compile the project and obtain the .hex file.

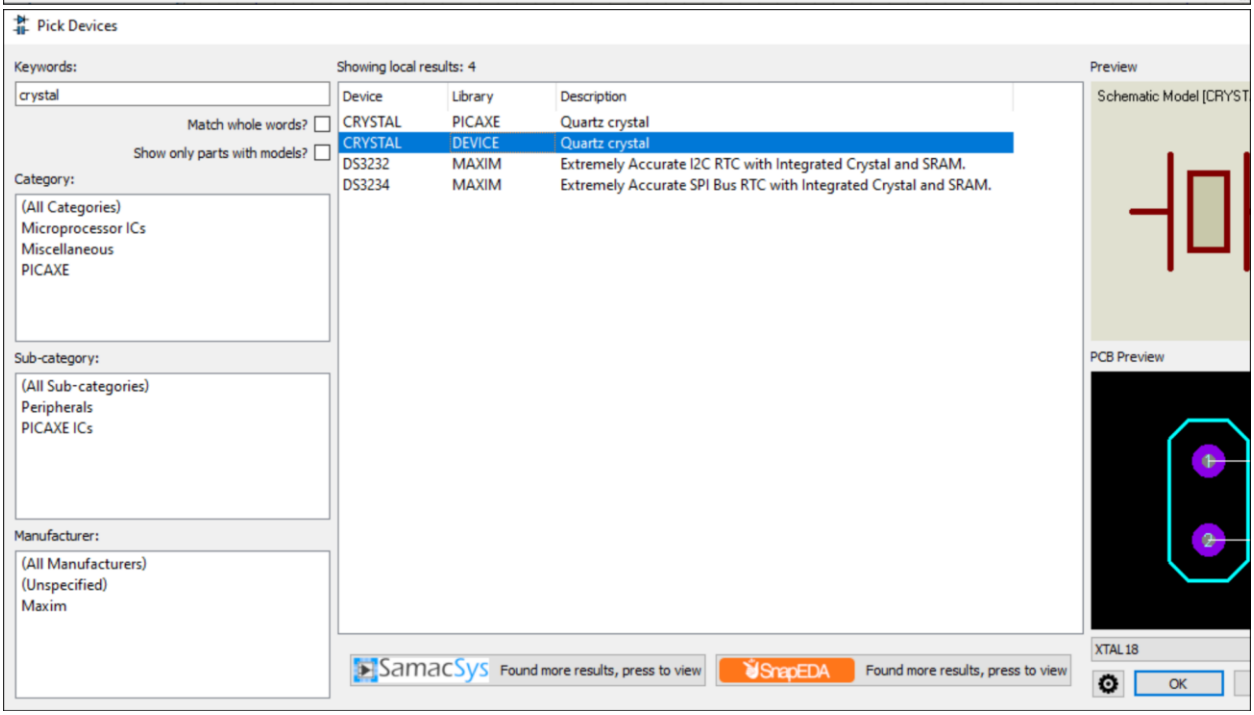


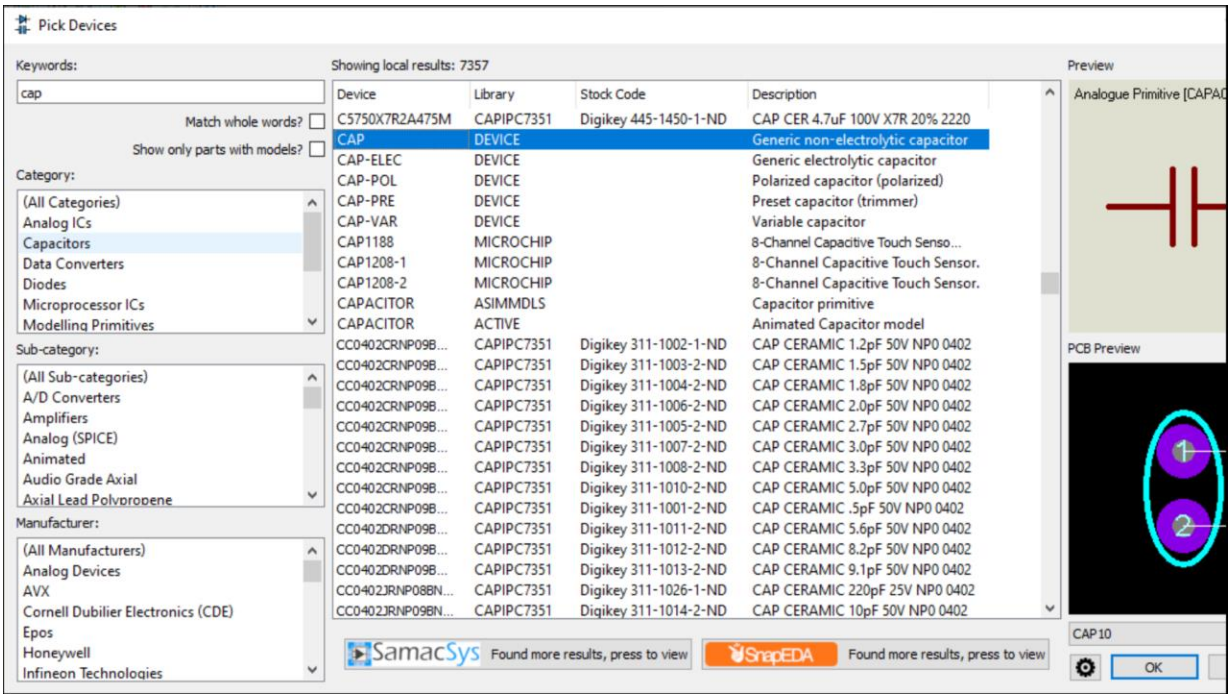
6. Add the ATmega32, two DC motors, an L293D motor driver, a 16 MHz crystal, two 22 pF capacitors, a 10 kΩ resistor, 5 V and 12 V power terminals, and GND to the schematic. Connect PB0–PB3 to the L293D's IN1–IN4 pins, and connect the motors to its output pins.



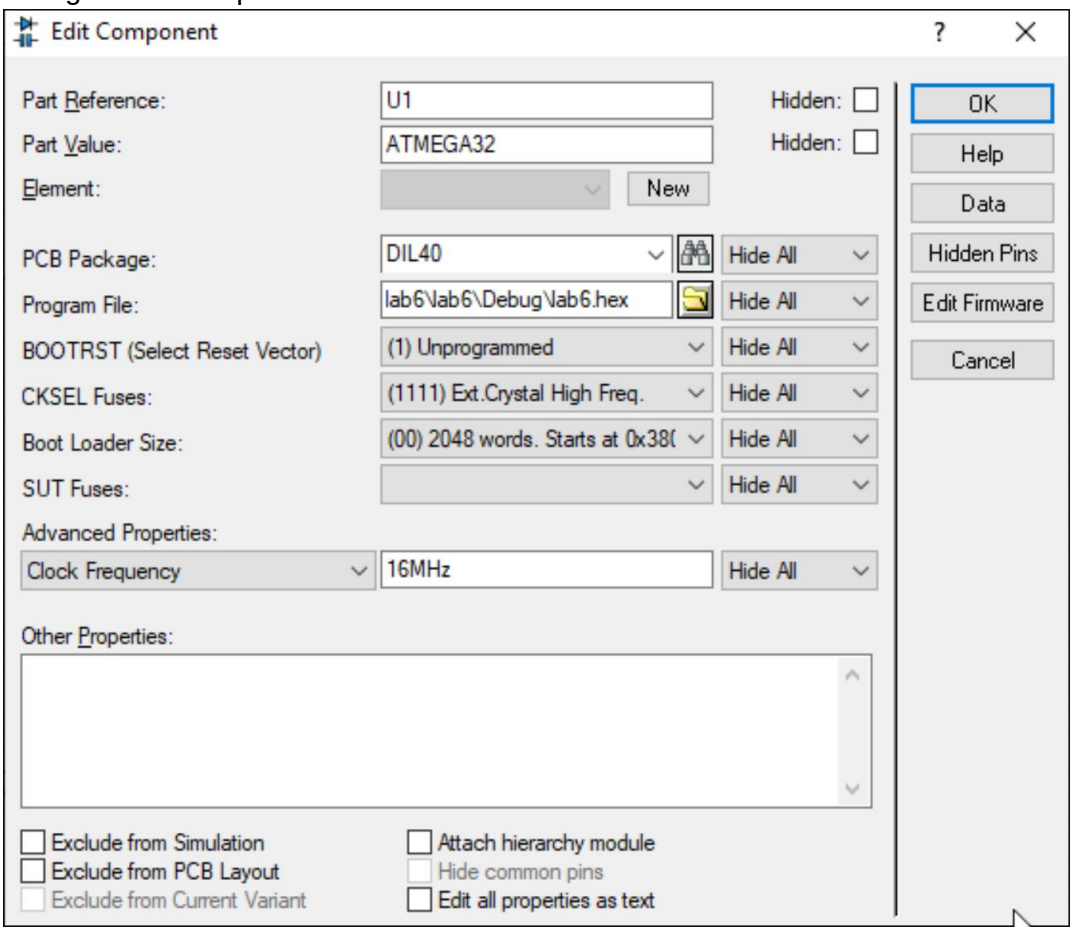



Pick Devices
Keywords:
dc motor
Match whole words?
Show only parts with models?
Category:
(All Categories)
Analog ICs
Electromechanical
Mechanics
Sub-category:
(All Sub-categories)
Miscellaneous
Manufacturer:
(All Manufacturers)
(Unspecified)
Texas Instruments
Toshiba
Showing local results: 11
Device
Library
Description
BLDC-STAR MOTORS
BLDC-TRIANGLE MOTORS
DC-GENERATOR MOTORS Animated DC Generator (Dynamo).
DRV8871 ANALOG 3.6-A Brushed DC Motor Driver With Internal Current Sen
FAN-DC MOTORS
MOTOR ACTIVE Simple DC Motor model
MOTOR MOTORS Simple DC Motor model
MOTOR-BLDCM MOTORS Animated Brushless DC Motor model
MOTOR-DC MOTORS Animated DC Motor model With Inertia And Loading
MOTOR-ENCODER MOTORS Animated DC Motor model With Inertia, Loading, and pc
TB6612FNG ANALOG Driver IC for Dual DC motor.
SnapEDA
Found more results, press to view
Preview
Schematic Model [DCMOTOR]
+88.8
PCB Preview
OK
Cancel

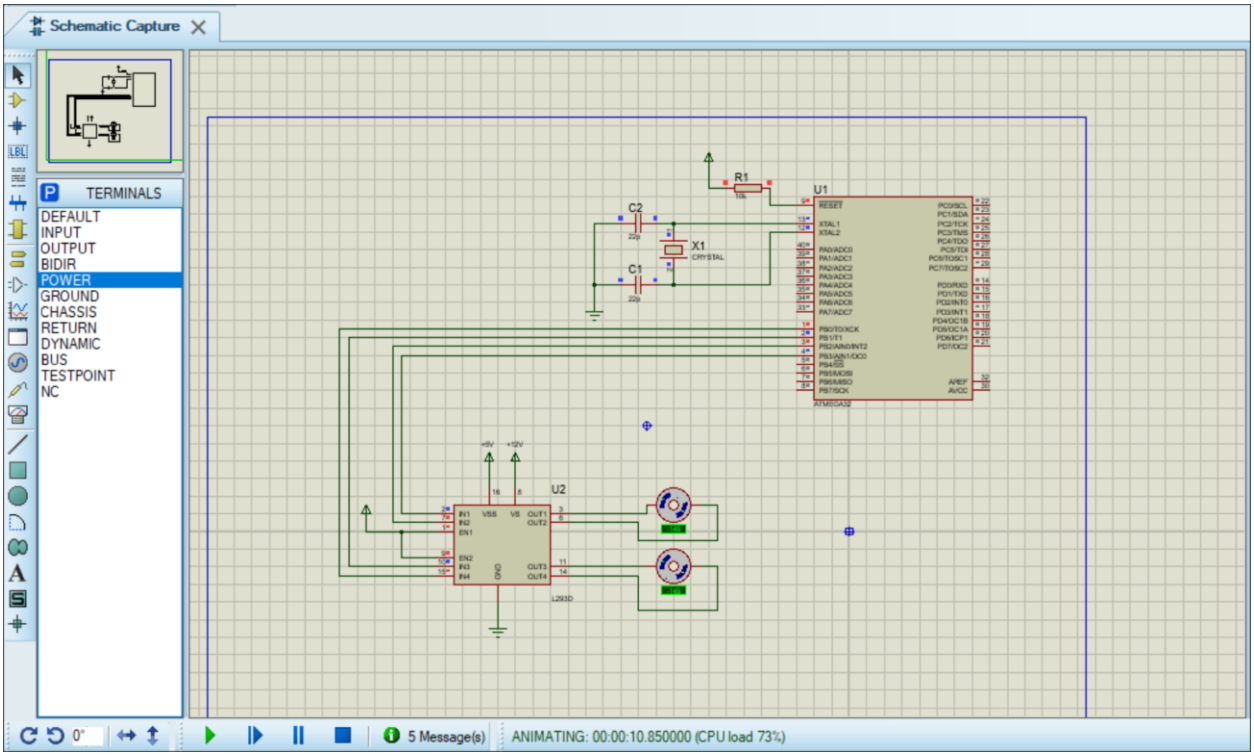

Pick Devices
Keywords:
crystal
Match whole words?
Show only parts with models?
Category:
(All Categories)
Microprocessor ICs
Miscellaneous
PICAXE
Sub-category:
(All Sub-categories)
Peripherals
PICAXE ICs
Manufacturer:
(All Manufacturers)
(Unspecified)
Maxim
Showing local results: 4
Device
Library
Description
CRYSTAL PICAXE Quartz crystal
CRYSTAL DEVICE Quartz crystal
DS3232 MAXIM Extremely Accurate I2C RTC with Integrated Crystal and SRAM.
DS3234 MAXIM Extremely Accurate SPI Bus RTC with Integrated Crystal and SRAM.
SamacSys
Found more results, press to view
SnapEDA
Found more results, press to view
Preview
Schematic Model [CRYST
PCB Preview
XTAL 18
OK



7. Load the compiled .hex file into the microcontroller's properties (Program File) and configure the component



8. Run the simulation and verify that both DC motors rotate clockwise and anticlockwise, then stop, according to the program.



Conclusion:

In this laboratory work, I programmed the ATmega32 to control two DC motors through the L293D. The motors changed direction and stopped according to the program.